

NatureBank AI — The Intelligence Layer for Biodiversity Credits & Natural Capital Markets

The Vision

NatureBank AI is the operating system for the **\$200B+ natural capital economy** — providing the intelligence infrastructure to measure, verify, trade, and manage biodiversity credits at planetary scale.

Just as carbon markets needed Verra and Gold Standard to create trust and liquidity, the emerging biodiversity credit market needs an AI-native intelligence layer. NatureBank AI is the Stripe + Bloomberg Terminal for nature finance.

The Problem

The Biodiversity Crisis Meets Finance

- **\$44 trillion of economic value** depends on nature (World Economic Forum)
- **Global biodiversity is collapsing** — 1 million species face extinction
- **Regulations are forcing action:** EU CSRD, TNFD framework, CBD Global Biodiversity Framework
- **But there’s no infrastructure:** No standardized measurement, no trusted verification, no liquid markets

Why Current Solutions Fail

Problem	Current State	Impact
Measurement chaos	50+ competing standards, inconsistent methodologies	Can’t compare or value biodiversity assets
Verification bottleneck	Manual field audits, takes 6-18 months	Projects stall, fraud goes undetected
Market fragmentation	Bilateral deals, no price discovery	Illiquid, inefficient, low trust
Data silos	Satellite, acoustic, eDNA data not integrated	Incomplete picture of ecosystem health
Corporate confusion	CFOs can’t quantify nature-related risks	Compliance failures, greenwashing accusations

The Market Gap

- **Carbon markets:** \$2B+ → mature infrastructure
- **Biodiversity markets:** <\$100M → no infrastructure
- But biodiversity will be **10x larger** because it’s more complex, more localized, and now mandatory

The Solution: NatureBank AI

Core Platform Components

1. BiomeOS™ — Multi-Modal Ecosystem Intelligence Continuous, AI-powered biodiversity monitoring combining: - **Satellite imagery** — Land cover, habitat connectivity, deforestation alerts - **Acoustic monitoring** — Species detection via bioacoustic AI (birds, mammals, insects) - **Environmental DNA (eDNA)** — Water/soil sampling for species presence - **Camera traps** — Visual species verification with computer vision - **Citizen science data** — iNaturalist, eBird integration - **IoT sensors** — Soil health, water quality, microclimate

Output: Real-time Biodiversity Health Score™ for any location on Earth

2. CreditForge™ — Automated Credit Origination End-to-end biodiversity credit creation: - **Baseline assessment** — AI-generated biodiversity baseline using historical data - **Additionality modeling** — Prove conservation impact beyond business-as-usual - **Methodology selection** — Automatically match project to appropriate standard (Wallacea Trust, Plan Vivo, Verra NCS) - **Documentation generation** — Auto-generate Project Design Documents - **Permanence insurance** — Integrated buffer pools and insurance products

Result: Reduce credit origination from 18 months to 90 days

3. VerifyNet™ — AI-Powered Verification Continuous monitoring replaces point-in-time audits: - **Anomaly detection** — Flag ecosystem degradation in real-time - **Remote verification** — 90% of verification via satellite + sensors - **Auditor marketplace** — Certified third-party verifiers for ground-truthing - **Blockchain registry** — Immutable credit ledger with full provenance - **Fraud detection** — Pattern recognition for double-counting, over-crediting

Outcome: Verification costs down 80%, integrity up 10x

4. NatureExchange™ — The Trading Platform Liquid marketplace for biodiversity credits: - **Standardized products** — Biodiversity Unit (BDU) as universal measure - **Price discovery** — Transparent bid/ask, reference pricing - **Corporate procurement** — Bulk purchasing for TNFD compliance - **Retail participation** — Consumer-facing “protect a hectare” products - **Derivative products** — Futures, options for hedging - **Impact tokens** — Tradeable impact certificates for specific projects

5. ComplianceOS™ — Corporate Nature Intelligence TNFD/CSRD compliance automation: - **Nature dependency mapping** — Link supply chain to biodiversity hotspots - **Risk quantification** — Financial exposure to ecosystem degradation - **Target setting** — Science-based targets for nature - **Reporting automation** — TNFD-aligned disclosures, audit-ready - **Portfolio optimization** — Maximize impact per dollar of nature investment

How It Works

For Landowners & Conservation Projects

1. REGISTER → Submit land parcel, project details
2. ASSESS → AI analyzes baseline biodiversity (satellite + eDNA + acoustic)

3. PLAN → Receive conservation/restoration roadmap
4. IMPLEMENT → Execute project with ongoing monitoring
5. VERIFY → Continuous AI verification + periodic ground-truth
6. ISSUE → Credits minted to blockchain registry
7. SELL → List on NatureExchange or direct corporate deals
8. MONITOR → Ongoing compliance, permanence tracking

For Corporate Buyers

1. ASSESS → Map nature dependencies and risks across operations
2. SET TARGETS → Science-based nature targets aligned to GBF
3. PROCURE → Purchase verified credits via NatureExchange
4. INTEGRATE → Link credits to TNFD disclosures
5. REPORT → Auto-generated compliance reporting
6. COMMUNICATE → Credible impact stories backed by data

Business Model

Revenue Streams

Stream	Model	Projected Revenue (Y5)
Platform SaaS	\$50K-\$500K/year per enterprise	\$80M
Credit transaction fees	3-5% of traded value	\$120M
Verification services	\$5K-\$50K per project	\$40M
Data & analytics	API access, custom reports	\$30M
Insurance products	Permanence guarantees	\$20M
White-label platform	Government/registry licensing	\$10M

Year 5 ARR Target: \$300M+

Unit Economics

- **CAC:** \$25K (enterprise sales)
- **LTV:** \$500K+ (multi-year platform + transactions)
- **LTV:CAC:** 20:1
- **Gross margin:** 75%+ (software + data)

Market Opportunity

Total Addressable Market (TAM)

- **Natural capital markets:** \$200B by 2030 (projected)
- **TNFD compliance spend:** \$50B annually by 2028
- **Biodiversity credit trading:** \$25B by 2030

Serviceable Market (SAM)

- **Enterprise biodiversity intelligence:** \$20B
- **Credit infrastructure fees:** \$5B
- **Verification services:** \$2B

Current Market Timing

Catalyst	Impact	Timeline
EU CSRD mandatory reporting	50,000 companies need nature data	2025-2026
TNFD adoption acceleration	Mainstream corporate adoption	2026+
CBD COP16 outcomes	Binding national targets	2025
First scaled biodiversity credit markets	Proof of concept	2026-2027
Insurance industry integration	Nature risk pricing	2027+

We are in the 2008 moment for biodiversity what carbon was in 2008 — infrastructure being built, regulation forcing adoption, but no dominant platform.

Competitive Landscape

Why We Win

Competitor	Approach	Weakness	Our Advantage
Verra/Gold Standard	Traditional registries	Slow, manual, carbon-focused	AI-native, built for biodiversity
Planet/Satellologic	Satellite imagery	Data only, no marketplace	Full-stack platform
Pachama	Carbon focus	Limited biodiversity	Biodiversity-first design
Sylvera	Carbon ratings	No origination/trading	End-to-end platform
Restor/Google	Mapping tools	No commercial infrastructure	Marketplace + compliance

Moat Construction

1. **Data network effects** — More projects → better AI models → more accurate verification
2. **Liquidity network effects** — More buyers + sellers → better prices → more participants
3. **Regulatory relationships** — First-mover with CBD, TNFD, EU regulators
4. **Science partnerships** — Exclusive methodologies from top conservation science
5. **Enterprise lock-in** — Integration with corporate systems, compliance data

Go-To-Market Strategy

Phase 1: Foundation (Months 1-12)

Focus: Build credibility, prove technology

- Partner with 3-5 flagship conservation organizations
- Launch BiomeOS for 10 pilot regions
- Publish “State of Biodiversity Markets” report
- Secure methodology approval from one major standard
- **Target:** 50 registered projects, 100K hectares monitored

Phase 2: Expansion (Months 12-24)

Focus: Scale platform, launch marketplace

- Open NatureExchange beta with institutional buyers
- Expand monitoring to 50 countries
- Launch ComplianceOS for TNFD early adopters
- Build auditor network (100+ certified verifiers)
- **Target:** \$50M in credits traded, 500 projects, 5M hectares

Phase 3: Dominance (Months 24-36)

Focus: Become the standard

- Full global coverage
- White-label to national registries
- Launch biodiversity derivatives
- Expand to marine ecosystems
- **Target:** \$500M+ traded, 5,000 projects, 50M hectares

Initial Customer Segments

Segment	Pain Point	Entry Product	Deal Size
Consumer goods cos	Deforestation-free supply chains	ComplianceOS	\$200K-\$1M/yr
Financial institutions	Portfolio nature risk	Risk analytics	\$100K-\$500K/yr
Conservation NGOs	Project funding, verification	CreditForge	Rev share
Indigenous land trusts	Monetize conservation	Full platform	Rev share
Governments	National biodiversity accounting	White-label	\$1M-\$10M

Technology Architecture

AI/ML Stack

NatureBank AI Platform

INTELLIGENCE LAYER

Species Detection Models (Computer Vision, Bioacoustic)
Ecosystem Health Scoring (Multi-modal fusion)
Change Detection (Time-series satellite analysis)
Fraud Detection (Anomaly detection, pattern matching)
Credit Valuation Models (Pricing, risk assessment)

DATA LAYER

Satellite Data Pipeline (Sentinel, Planet, Maxar)
Acoustic Data Lake (Continuous audio ingestion)
eDNA Analysis Pipeline (Sequence matching, abundance)
Ground Truth Database (Field observations, camera)
Market Data (Pricing, transactions, inventory)

INFRASTRUCTURE LAYER

Blockchain Registry (Polygon, carbon-neutral)
API Gateway (Third-party data access)
Compliance Engine (TNFD, CSRD, GBF mapping)
Trading Infrastructure (Matching engine, settlement)

Key Technical Differentiators

1. **Multi-modal fusion** — Combine satellite, acoustic, eDNA, camera for holistic view
 2. **Continuous monitoring** — Real-time vs. point-in-time verification
 3. **Biodiversity-specific AI** — Not repurposed carbon models
 4. **Planetary scale** — Architecture for monitoring any ecosystem on Earth
 5. **Financial-grade infrastructure** — Exchange-quality trading systems
-

Traction & Validation

Early Indicators

- **5 LOIs** from Fortune 500 consumer goods companies
- **Partnership discussions** with IUCN, WWF, Conservation International
- **Methodology collaboration** with Cambridge Conservation Initiative
- **Pilot deployment** in Amazon, Congo Basin, Southeast Asia
- **Advisory board** including former TNFD taskforce members

Key Milestones Ahead

Milestone	Target Date	Significance
First verified biodiversity credit	Q2 2026	Proof of technology
\$10M in credits traded	Q4 2026	Market validation
TNFD reporting partnership	Q2 2027	Enterprise channel
1M hectares monitored	Q4 2027	Scale demonstration
\$100M ARR	Q4 2028	Business validation

Team Requirements

Core Founding Team

- **CEO** — Deep climate/nature markets experience, fundraising track record
- **CTO** — Remote sensing/ML background, scaled data platforms
- **Chief Science Officer** — Conservation biology PhD, credibility with science community
- **Chief Commercial Officer** — Enterprise SaaS sales, sustainability buyer relationships

Key Early Hires

- Conservation finance specialists
- Remote sensing engineers
- Bioacoustic ML researchers
- Blockchain/registry developers
- Regulatory affairs (EU, US, UK)

Advisory Needs

- Former TNFD taskforce members
- Indigenous rights experts
- Commodity trading veterans
- Conservation organization leaders

Funding Strategy

Seed Round: \$8M

Use of funds: - Core platform development (60%) - Pilot deployments in 5 regions (20%) - Science partnerships and methodology (15%) - Team (5 founding + 10 early hires)

Series A: \$35M (Month 18)

Milestones required: - 100+ registered projects - \$10M in credits originated - 3+ enterprise customers - Methodology approval from major standard

Series B: \$100M (Month 36)

Milestones required: - \$50M ARR - Market-leading position in 2+ regions - Government partnerships - Profitable unit economics proven

Risk Factors & Mitigation

Risk	Probability	Impact	Mitigation
Regulatory uncertainty	Medium	High	Active engagement with policymakers, flexible platform
Market adoption slower than expected	Medium	High	Compliance-driven entry, not voluntary-only
Technology risk (AI accuracy)	Low	High	Multi-modal redundancy, human-in-loop verification
Competition from incumbents	Medium	Medium	Speed, technology moat, network effects
Greenwashing backlash	Medium	High	Radical transparency, conservative claims
Indigenous rights concerns	Medium	High	Indigenous-led governance, benefit sharing

Why Now?

Regulatory Forcing Function

- **CSRD**: 50,000 EU companies must report nature impacts starting 2025
- **TNFD**: Voluntary now → expected mandatory by 2028
- **SEC climate rules**: Nature next after carbon
- **CBD GBF**: Nations committed to 30x30, need measurement

Technology Readiness

- **Satellite resolution**: 3m resolution at <\$1/km²
- **Bioacoustic AI**: Species detection accuracy >95%
- **eDNA**: Costs dropped 90% in 5 years
- **Foundation models**: Generalizable ecosystem understanding

Market Momentum

- **First biodiversity credits** traded in 2024
- **Wallacea Trust** launched first scalable methodology
- **Corporate demand** exceeds supply 100:1
- **Financial institutions** building nature practices

The Billion-Dollar Path

Year 1: \$3M ARR

- Foundation laid, technology proven
- First credits traded

Year 3: \$50M ARR

- Market leader in 2 regions
- 1,000+ projects monitored
- Enterprise product launched

Year 5: \$300M ARR

- Global platform
- \$1B+ in credits traded annually
- Government partnerships
- Clear path to profitability

Year 7: \$1B+ valuation path

- Market standard for biodiversity intelligence
 - Acquisition target for: Bloomberg, S&P, Salesforce, Microsoft
 - IPO candidate
-

The Ask

Raising \$8M Seed to: 1. Build core BiomeOS and CreditForge platform 2. Deploy pilots in Amazon, Congo Basin, Southeast Asia 3. Secure methodology partnerships 4. Hire founding team of 15

Looking for investors who: - Understand climate/nature markets - Have enterprise software experience - Bring strategic value (corporate relationships, policy access) - Are patient for market timing

Executive Summary

NatureBank AI is building the intelligence infrastructure for the \$200B natural capital economy. As the world shifts from carbon-only to nature-positive, companies face mandatory biodiversity disclosure with no tools to comply. We provide the full-stack platform: AI-powered monitoring, automated credit origination, continuous verification, and liquid trading infrastructure.

The opportunity: Be the Bloomberg + Stripe of biodiversity markets — a platform that becomes essential infrastructure as the market scales 100x over the next decade.

The team: Conservation science + fintech + enterprise software.

The timing: Perfect. Regulation forcing adoption, technology ready, market forming, no dominant player.

Join us in building the financial infrastructure to save nature at scale.

“What gets measured gets managed. What gets traded gets conserved.”

NatureBank AI — The Intelligence Layer for Natural Capital

Contact

Website: naturebank.ai (*coming soon*) **Email:** founders@naturebank.ai **Location:** San Francisco / London / Singapore